Namespace Solstice.Scheduled.Core

Classes

CoreBackgroundService

Class CoreBackgroundService

Namespace: Solstice.Scheduled.Core

Assembly: Solstice.Scheduled.dll

```
public abstract class CoreBackgroundService : IHostedService
```

Inheritance

object ← CoreBackgroundService

Implements

IHostedService

Derived

CoreQueuedBackgroundService, CoreScopedProcessor<TDbContext>

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Methods

ExecuteAsync(CancellationToken)

```
protected virtual Task ExecuteAsync(CancellationToken cancellationToken)
```

Parameters

`cancellationToken` CancellationToken

Returns

Task

ProcessAsync(CancellationToken)

```
protected abstract Task ProcessAsync(CancellationToken cancellationToken)
```

Parameters

`cancellationToken` CancellationToken

Returns

Task

StartAsync(CancellationToken)

Triggered when the application host is ready to start the service.

```
public Task StartAsync(CancellationToken cancellationToken)
```

Parameters

`cancellationToken` CancellationToken

Indicates that the start process has been aborted.

Returns

Task

A Task that represents the asynchronous Start operation.

StopAsync(CancellationToken)

Triggered when the application host is performing a graceful shutdown.

```
public Task StopAsync(CancellationToken cancellationToken)
```

Parameters

`cancellationToken` CancellationToken

Indicates that the shutdown process should no longer be graceful.

Returns

Task

A Task that represents the asynchronous Stop operation.

Namespace Solstice.Scheduled.Injections

Classes

ScheduledInjections

Class ScheduledInjections

Namespace: Solstice.Scheduled.Injections

Assembly: Solstice.Scheduled.dll

```
public static class ScheduledInjections
```

Inheritance

object ← ScheduledInjections

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Methods

AddCronServices(IServiceCollection, params Type[])

Injector to add hosted services to the services

```
public static void AddCronServices(this IServiceCollection services, params
Type[] types)
```

Parameters

`services` IServiceCollection

`types` Type[]

Remarks

services.AddCronServices(typeof(HostedService1), typeof(HostedService2), ...)

Exceptions

ArgumentException

AddQueue(IServiceCollection, int)

Injector to add a queue to the services

```
public static void AddQueue(this IServiceCollection services, int capacity = 100000)
```

Parameters

`services` IServiceCollection

`capacity` int

Queue capacity

Namespace Solstice.Scheduled.Queue

Classes

CoreQueuedBackgroundService

DefaultBackgroundTaskQueue

QueueItem

Interfaces

IBackgroundTaskQueue

Class CoreQueuedBackgroundService

Namespace: Solstice.Scheduled.Queue

Assembly: Solstice.Scheduled.dll

```
public sealed class CoreQueuedBackgroundService : CoreBackgroundService,
IHostedService
```

Inheritance

object ← CoreBackgroundService ← CoreQueuedBackgroundService

Implements

IHostedService

Inherited Members

CoreBackgroundService.StartAsync(CancellationToken) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

CoreQueuedBackgroundService(IBackgroundTaskQueue, ILogger<CoreQueuedBackgroundService>)

```
public CoreQueuedBackgroundService(IBackgroundTaskQueue taskQueue,
ILogger<CoreQueuedBackgroundService> logger)
```

Parameters

`taskQueue` IBackgroundTaskQueue

`logger` ILogger<CoreQueuedBackgroundService>

Methods

ExecuteAsync(CancellationToken)

```
protected override Task ExecuteAsync(CancellationToken stoppingToken)
```

Parameters

`stoppingToken` CancellationToken

Returns

Task

ProcessAsync(CancellationToken)

```
protected override Task ProcessAsync(CancellationToken stoppingToken)
```

Parameters

`stoppingToken` CancellationToken

Returns

Task

StopAsync(CancellationToken)

```
public Task StopAsync(CancellationToken stoppingToken)
```

Parameters

`stoppingToken` CancellationToken

Returns

Task

Class DefaultBackgroundTaskQueue

Namespace: Solstice.Scheduled.Queue

Assembly: Solstice.Scheduled.dll

```
public sealed class DefaultBackgroundTaskQueue : IBackgroundTaskQueue
```

Inheritance

object ← DefaultBackgroundTaskQueue

Implements

IBackgroundTaskQueue

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

DefaultBackgroundTaskQueue(int)

```
public DefaultBackgroundTaskQueue(int capacity)
```

Parameters

`capacity` int

Methods

ClearChannel()

```
public void ClearChannel()
```

DequeueAsync(CancellationToken)

```
public ValueTask<QueueItem> DequeueAsync(CancellationToken cancellationToken)
```

Parameters

`cancellationToken` CancellationToken

Returns

ValueTask<QueueItem>

QueueBackgroundWorkItemAsync(QueueItem)

```
public ValueTask QueueBackgroundWorkItemAsync(QueueItem workItem)
```

Parameters

`workItem` QueueItem

Returns

ValueTask

Interface IBackgroundTaskQueue

Namespace: Solstice.Scheduled.Queue

Assembly: Solstice.Scheduled.dll

```
public interface IBackgroundTaskQueue
```

Methods

ClearChannel()

```
void ClearChannel()
```

DequeueAsync(CancellationToken)

```
ValueTask<QueueItem> DequeueAsync(CancellationToken cancellationToken)
```

Parameters

`cancellationToken` CancellationToken

Returns

ValueTask<QueueItem>

QueueBackgroundWorkItemAsync(QueueItem)

```
ValueTask QueueBackgroundWorkItemAsync(QueueItem workItem)
```

Parameters

`workItem` QueueItem

Returns

ValueTask

Class QueueItem

Namespace: Solstice.Scheduled.Queue

Assembly: Solstice.Scheduled.dll

```
public class QueueItem
```

Inheritance

object ← QueueItem

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Properties

Guid

```
public Guid Guid { get; init; }
```

Property Value

Guid

WorkItem

```
public Func<CancellationToken, Task>? WorkItem { get; init; }
```

Property Value

Func<CancellationToken, Task>

Namespace Solstice.Scheduled.Scheduled

Classes

CoreScheduledProcessor<TDbContext>

CoreScopedProcessor<TDbContext>

Class CoreScheduledProcessor<TDbContext>

Namespace: Solstice.Scheduled.Scheduled

Assembly: Solstice.Scheduled.dll

```
public abstract class CoreScheduledProcessor<TDbContext> :
CoreScopedProcessor<TDbContext>, IHostedService where TDbContext : DbContext
```

Type Parameters

`TDbContext`

Inheritance

object ← CoreBackgroundService ← CoreScopedProcessor<TDbContext> ← CoreScheduledProcessor<TDbContext>

Implements

IHostedService

Inherited Members

CoreScopedProcessor<TDbContext>._serviceScopeFactory , CoreScopedProcessor<TDbContext>.ProcessAsync(CancellationToken) , CoreScopedProcessor<TDbContext>.ProcessInScopeAsync(TDbContext, CancellationToken) , CoreBackgroundService.StartAsync(CancellationToken) , CoreBackgroundService.StopAsync(CancellationToken) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

CoreScheduledProcessor(IServiceScopeFactory)

```
protected CoreScheduledProcessor(IServiceScopeFactory serviceScopeFactory)
```

Parameters

`serviceScopeFactory` IServiceScopeFactory

Properties

Schedule

```
protected abstract string Schedule { get; }
```

Property Value

string

Methods

ExecuteAsync(CancellationToken)

```
protected override Task ExecuteAsync(CancellationToken stoppingToken)
```

Parameters

`stoppingToken` CancellationToken

Returns

Task

Class CoreScopedProcessor<TDbContext>

Namespace: Solstice.Scheduled.Scheduled

Assembly: Solstice.Scheduled.dll

```
public abstract class CoreScopedProcessor<TDbContext> : CoreBackgroundService,
IHostedService where TDbContext : DbContext
```

Type Parameters

`TDbContext`

Inheritance

object ← CoreBackgroundService ← CoreScopedProcessor<TDbContext>

Implements

IHostedService

Derived

CoreScheduledProcessor<TDbContext>

Inherited Members

CoreBackgroundService.StartAsync(CancellationToken) , CoreBackgroundService.StopAsync(CancellationToken) , CoreBackgroundService.ExecuteAsync(CancellationToken) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

CoreScopedProcessor(IServiceScopeFactory)

```
protected CoreScopedProcessor(IServiceScopeFactory serviceScopeFactory)
```

Parameters

`serviceScopeFactory` IServiceScopeFactory

Fields

_serviceScopeFactory

```
protected readonly IServiceScopeFactory _serviceScopeFactory
```

Field Value

IServiceScopeFactory

Methods

ProcessAsync(CancellationToken)

```
protected override Task ProcessAsync(CancellationToken cancellationToken)
```

Parameters

`cancellationToken` CancellationToken

Returns

Task

ProcessInScopeAsync(TDbContext, CancellationToken)

```
public abstract Task ProcessInScopeAsync(TDbContext dbContext,
CancellationToken cancellationToken)
```

Parameters

`dbContext` TDbContext

`cancellationToken` CancellationToken

Returns

Task